Yantian Zha

Assistant Professor, Department of Computer Science

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RESEARCH PROFILE

Embodied artificial intelligence and robotics, with emphasis on robot learning, computer vision, planning, reinforcement and imitation learning, human-robot interaction, foundation models, simulation, and perception and control for intelligent systems operating across air, water, and land.

ACADEMIC APPOINTMENTS

Assistant Professor, Department of Computer Science **Aug. 2025-Present**
North Carolina A&T State University
Tenure-track appointment; Director, Omnidomain AI and Robotics Laboratory.
Courses: COMP 645 Artificial Intelligence, Fall 2025 and Spring 2026.

Adjunct Lecturer **Aug. 2023-Present**
University of Maryland, College Park
Courses include CMSC 421 Introduction to Artificial Intelligence, CMSC 848J Cognitive Robotics, and ENPM 808Z Cognitive Robotics.

Postdoctoral Associate **July 2024-Aug. 2025**
University of Maryland, College Park
Supervisors: Professors Balakumar Balachandran and Miao Yu.

Postdoctoral Associate **July 2022-Aug. 2023**
University of Maryland, College Park
Supervisor: Professor Yiannis Aloimonos.

Robotics Research Intern **Mar.-Aug. 2020**
ABB Corporate Research Center, United States
Mentor: Dr. Jianjun Wang. Conducted robotics research and contributed software subsequently transferred to a product team.

Teaching Assistant **Aug.-Dec. 2016**
Arizona State University
Course: Introduction to Artificial Intelligence.

EDUCATION

Ph.D. in Computer Science, Artificial Intelligence **2022**
Arizona State University, Tempe, Arizona
2017-2022. Research advisor: Professor Subbarao Kambhampati.

M.S. in Computer Engineering, Computer Systems **2017**
Arizona State University, Tempe, Arizona
2015-2017. Research advisor: Professor Subbarao Kambhampati.

B.E. in Electronics Engineering, Automation **2014**
Southeast University Chengxian College, China
2010-2014. Research advisor: Professor Xudong Ma.

SPONSORED RESEARCH AWARDS AND SUPPORT

1. ERI: Towards Sustainable Aquaculture: Video-Based Reconstruction of Digital Underwater Animals for Robot Learning of Aquatic Capture Tasks. National Science Foundation, $200,000, July 1, 2026-June 30, 2028. Principal Investigator. Awarded 2026.

2. Startup Package. North Carolina A&T State University, $110,000, July 1, 2026-June 30, 2028. Principal Investigator.

OTHER SPONSORED RESEARCH EXPERIENCE

- **Planning-and-Plan-Recognition-Driven Attention Models for Supporting Active Perception in Decentralized Multi-Agent Systems.** Office of Naval Research, 2018. Principal Investigator: Subbarao Kambhampati; Co-Principal Investigator: Baoxin Li. Contributing researcher.

PUBLICATIONS

Publications are grouped by type and listed in reverse chronological order. **Yantian Zha** is bolded; an asterisk denotes equal contribution or an equal advising role, as indicated.

Peer-Reviewed Journal Articles

1. Zhuo, H. H., **Zha, Y.**, and Kambhampati, S. (2019). Discovering Underlying Plans Based on Shallow Models. ACM Transactions on Intelligent Systems and Technology.

Peer-Reviewed Conference Papers

1. Shrestha, S., **Zha, Y.***, Banagiri, S. N., Gao, G.*, Aloimonos, Y.*, and Fermuller, C.* (2025). NatSGLD: A Dataset with Speech, Gesture, Logic, and Demonstration for Robot Learning in Natural Human-Robot Interaction. ACM/IEEE International Conference on Human-Robot Interaction, 1093-1098. *Equal advising role, as indicated in the source CV.
2. Guan, L., Zhou, Y., Liu, D., **Zha, Y.**, Ben Amor, H., and Kambhampati, S. (2024). Task Success Is Not Enough: Investigating the Use of Video-Language Models as Behavior Critics for Catching Undesirable Agent Behaviors. Conference on Language Modeling. OpenReview.
3. **Zha, Y.***, Guan, L.*, and Kambhampati, S. (2024). Learning from Ambiguous Demonstrations with Self-Explanation Guided Reinforcement Learning. Proceedings of the AAAI Conference on Artificial Intelligence, 38, 10395-10403. *Equal contribution.
4. Kambhampati, S., Sreedharan, S., Verma, M., **Zha, Y.**, and Guan, L. (2022). Symbols as a Lingua Franca for Bridging Human-AI Chasm for Explainable and Advisable AI Systems. Proceedings of the AAAI Conference on Artificial Intelligence, Senior Member Presentation Track.
5. **Zha, Y.**, Bhambri, S., and Guan, L. (2021). Contrastively Learning Visual Attention as Affordance Cues from Demonstrations for Robotic Grasping. IEEE/RSJ International Conference on Intelligent Robots and Systems, 7835-7842.
6. Kulkarni, A., **Zha, Y.**, Chakraborti, T., Vadlamudi, S. G., Zhang, Y., and Kambhampati, S. (2019). Explicability as Minimizing Distance from Expected Behavior. International Conference on Autonomous Agents and Multiagent Systems.
7. **Zha, Y.**, Li, Y., Gopalakrishnan, S., Li, B., and Kambhampati, S. (2018). Recognizing Plans by Learning Embeddings from Observed Action Distributions. Proceedings of the 17th International Conference on Autonomous Agents and Multiagent Systems, 2153-2155.

Workshop Papers

1. **Zha, Y.**, Li, Y., Yu, T., Kambhampati, S., and Li, B. (2019). Plan-Recognition-Driven Attention Modeling for Visual Recognition. AAAI Workshop on Plan, Activity, and Intent Recognition.

Preprints

1. Chen, H., Cheng, L., Jilan, T. Z., Fuller, I., Caesar, L., Wu, J., and **Zha, Y.** (2026). Video2SwimFish: An Automated Pipeline for Reconstructing Controllable Fish Models and Biological Locomotion from Real Fish Videos. arXiv preprint arXiv:2609.38966.
2. Yang, W.*, Kona, K.*, Jain, Y.*, Atzili, T.*, Bhamidipati, A., Lin, X., and **Zha, Y.** (2024). AAM-SEALS: Developing Aerial-Aquatic Manipulators in Sea, Air, and Land Simulator. arXiv preprint.

RESEARCH OUTPUTS

- Video2SwimFish preprint: arxiv.org/abs/2609.38966.
- AAM-SEALS simulation project: aam-seals.umd.edu.
- NatSGLD dataset and project materials: snehesh.com/natsgld.

INVITED AND CONFERENCE PRESENTATIONS

1. **Embodied and Human-Aligned Intelligence across Air, Water, and Land: From Simulation to Data-Efficient Learning.** Spring Research Symposium for an NSF-supported Data Science and AI project, North Carolina A&T State University, May 16, 2025. Invited keynote presentation.
2. **Cognitively-Enhanced Robotic Manipulation across Sea, Air, and Land.** Maryland Research Day, College Park, Maryland, August 2024. Invited presentation.
3. **Cognitively-Enhanced Robotic Manipulation across Sea, Air, and Land.** Robotics Institute, Carnegie Mellon University, Pittsburgh, Pennsylvania, July 2024. Invited presentation.
4. **Perceiving, Acting, Planning, and Self-Explaining: A Cognitive Quartet for Future Manufacturing.** Maryland Robotics Center Research Symposium, College Park, Maryland, May 2024. Invited presentation.
5. **Large Vision-Language Models for Self-Explanation-Guided Robot Learning.** University of Maryland, College Park, Maryland, November 2023. Invited presentation.
6. **Perceiving, Acting, Planning, and Self-Explaining: A Cognitive Quartet for Effective Human-Robot Collaboration.** University of Central Florida, Orlando, Florida, March 2023. Invited presentation.
7. **Contrastively Learning Visual Attention as Affordance Cues from Demonstrations for Robotic Grasping.** IEEE/RSJ International Conference on Intelligent Robots and Systems, virtual, September 2021. Conference presentation.
8. **Plan-Recognition-Driven Attention Modeling for Visual Recognition.** AAAI Workshop on Plan, Activity, and Intent Recognition, Honolulu, Hawaii, January 2019. Conference presentation.
9. **Recognizing Plans by Learning Embeddings from Observed Action Distributions.** ICML/IJCAI/AAMAS Workshop on Planning and Learning, Stockholm, Sweden, July 2018. Conference presentation.

HONORS AND AWARDS

- **Maryland Robotics Center Postdoctoral Fellowship**, University of Maryland, 2022-2023. Project: Self-Explanation-Guided Deep Reinforcement of Imitation Learning from Human Advice for Task-Driven Perception and Robotic Manipulation.

PROFESSIONAL AND UNIVERSITY SERVICE

Peer Review and Professional Service

- Reviewed two conference papers in 2025. Additional reviewer service has included IEEE Robotics and Automation Letters; Conference on Robot Learning; Robotics: Science and Systems; International Conference on Learning Representations; International Conference on Automated Planning and Scheduling; and IEEE Transactions on Image Processing.
- Workshop Organizer, Bridging the Gap between Cognitive Science and Robot Learning in the Real World: Progresses and New Directions, Conference on Robot Learning, November 2023.
- Invited Advisory Board member for a proposed human-computer interaction project at North Carolina A&T State University, 2026; participation contingent on project funding.
- Undergraduate Research Thesis Discussant, Gemstone Honors Program, University of Maryland, March 2024.

University and Outreach Service

- Member, AI Working Group, North Carolina A&T State University, 2025-Present. Contributed to the university AI strategy supporting the 2030 mission.
- Laboratory tours and robotics research demonstrations for visitors.

STUDENT MENTORING AND ADVISING

North Carolina A&T State University

- Hangong Chen, Ph.D. student, March 2025-Present.
- Lee Caesar, Ph.D. student, August 2025-Present.
- Tahsin Zaman, Ph.D. student, January 2026-Present.

University of Maryland, College Park

- Yangsheng, Ph.D. student, co-mentored, August 2024-Present.